ENSF-381: Assignment 2

Winter 2026

Department of Electrical and Software Engineering Schulich School of Engineering

Due: March 13, 2026, at 11:59 PM

Enhance the Ice Cream ordering website built in Assignment 1 by adding interactivity and dynamic features using JavaScript. Focus on DOM manipulation, form validations, API integration and real-time updates to create a responsive user experience.

Administrative Details

- The submission should be a ***single compressed file (.zip)*** that includes the complete source code and resources (e.g., images), ready for execution without modifications.
- Name the submitted file as: *Assignment2_ENSF381_Group#.*
  Example: Assignment2_ENSF381_Group01.zip
- Submit ONLY ONE copy.
- Include group members' names and UCIDs at the top of the index.html file as comments.

Instructions

- If you did not complete **Assignment 01**, a reference solution will be available on **March 02, 2026**, in **D2L** to help you proceed with this assignment.
- Concepts which will be needed but are not discussed in class: ***window.location.href, input.checked, setInterval***

Specifications

Implement the following features across the Ice Cream ordering website pages:

1. Menu View Page (menu_view.html) - 20 Marks

Requirements:

a). **Cart Section**:

Add a dedicated **Cart section** to menu_view.html with a clear heading (Cart) and a container where selected items are displayed.

- The Cart section must be positioned at the **bottom of the Menu View page content** (after the ice cream listing sections).
- The Cart section must be visible on the Menu View page at all times and update immediately when users add or remove items.
- If no items are selected, display an empty-state message (e.g., No items in cart.) inside the Cart section.

b). **Dynamic Cart Management**:

- Clicking "Add to Cart" or "Remove from Cart" on ice cream items updates the UI without reloading the page as follows:
    - When an item is added, a copy of the selected ice cream item must appear in the "Cart" section without reloading the page.
    - For each item shown in the Cart, display:
        - the item **name**,
        - the item **quantity** in parentheses beside the name (e.g., Peppermint Snow (2)),
        - and a **line total** on the far right equal to num_selected * item price (e.g., 2 * 5.99 = 11.98, displayed as $11.98).
    - When "Remove from Cart" is clicked, the item must be removed from the "Cart" section.

c). **Hover Effects**:

- Highlight ice cream item tiles on hover (e.g., change background color to #ffffff).

Seasonal Specials

Peppermint Snow

Price: $5.99

Add Remove

Caramel Apple Crunch

Price: $6.49

Add Remove

Maple Pecan Delight

Price: $6.29

Add Remove

Cart

No items in cart.

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Seasonal Specials

Peppermint Snow

Price: $5.99

Add Remove

Caramel Apple Crunch

Price: $6.49

Add Remove

Maple Pecan Delight

Price: $6.29

Add Remove

Cart

Caramel Apple Crunch (2)	$12.98
Maple Pecan Delight (1)	$6.29

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2. Customize Order Page (customize_order.html) - 30 Marks

Requirements:

a.) ***Order Timer:***

- Add an html element to display a 10-minute countdown timer in the MM:SS format. [*Use setInterval and element.textContent to implement the timer.*]. The timer must start automatically when the page loads.
- Place the timer at the **top of the Customize Order main content**, directly above the order form fields so it is always visible while customizing.
- Automatically redirect to *order_summary.html* when the timer expires. [*Use window.location.href for automatically redirecting to other pages.*]

b) ***Selection Validation****:*

- Validate that the user selects one base flavor (radio button required).
- At least one topping must be selected (checkbox required).
- If required selections are missing, display an alert message using JavaScript [*Use input.checked* to verify user selections before calculating price]

c) ***Price Calculation****:*

- Track selected options and update the total price when the user submits the order.
- The total price must be displayed on the page using JavaScript [Use element.textContent to update price display]
- Place the total price display **immediately before the "Finish Order" button** in the form layout.
- Base price = $6.
- Each selected topping adds $1.50 to total price.

Order Time Left: 06:55

Vanilla Dream

Choose Type:

- ○ Cup
- ○ Cone

Chocolate Bliss

Select toppings:

- ☐ Sprinkles
- ☐ Chocolate Chips
- ☐ Whipped Cream

Total Price: $6.00

Submit Order

3. Login Page (login.html) - 20 Marks

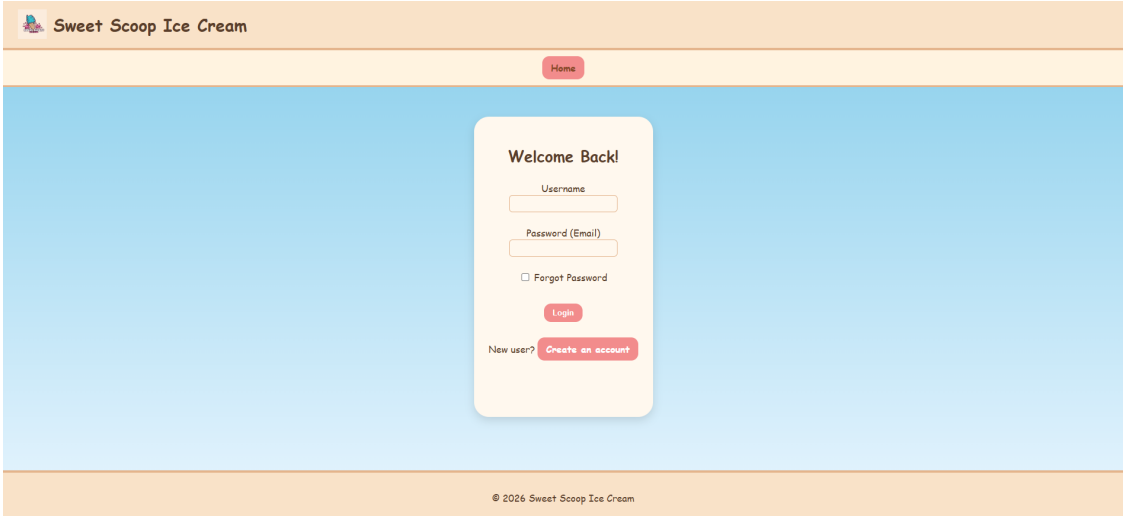
Requirements:

*a) **API Validation**:*

- Fetch user data from https://jsonplaceholder.typicode.com/users
- Validate credentials against the API's username and email fields.
- Username (case-insensitive) entered must match with any one of the usernames returned by the API.
- Password entered (case sensitive) must match with the corresponding users email address as returned by the API.
- Display success/error messages in a styled dynamic box.
- The styled dynamic message box must be displayed **beneath the Login form** (outside the <form> element, directly below it).
- In case of success, display the message for 2 seconds and then automatically load menu_view.html [*Use window.location.href for automatically redirecting to other pages.*]

*b) **Link to Signup Page**:*

Update the Login page to include a link for loading a Sign Up page for new users.



4. Signup Page (signup.html) - 30 Marks

Requirements:

a) ***Create signup.html***:

- Include a form with fields for username, password, confirm password, and email.
- Add a "Signup" button.
- Style the "Signup" button with a contrasting color of #4CAF50 (green), padding of 10px, border radius of 5px, margin of 10px, and opacity of 0.5.
- Add a hover effect by changing the background color to #45A049 (a slightly darker shade of green) and opacity of 1.0.
- The page should be in the same theme and layout as the Login page that was developed as part of Assignment 01.

b) ***Form Validation****:*

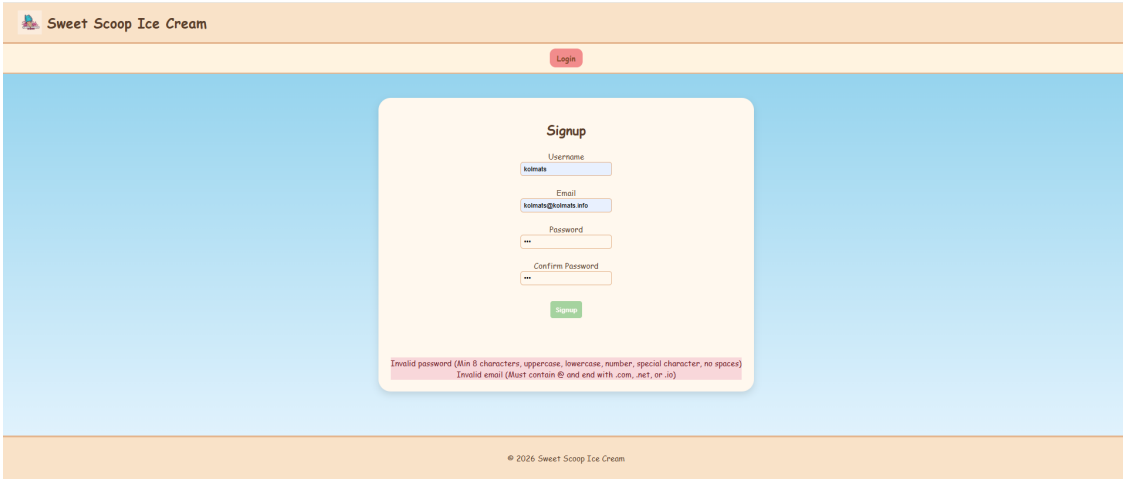
- Username:
  - Must be between 3 and 20 characters long.
  - Allowed characters: alphanumeric characters (letters A-Z, numbers 0-9), hyphens (-), and underscores (_).
  - Must start with a letter.
  - Cannot contain spaces or special characters other than hyphens and underscores.
- Password:
  - Must be at least 8 characters long.
  - Must contain at least one uppercase letter, one lowercase letter, one number, and one special character.
  - Allowed special characters: !@#$%^&*()-_=+[]{}|;:'",.<>?/`~.
  - Cannot contain spaces.
- Confirm Password:
  - Must match the password entered in the password field.
- Email:
  - Must be a valid email address format (e.g., username@example.com).
  - Cannot contain spaces.

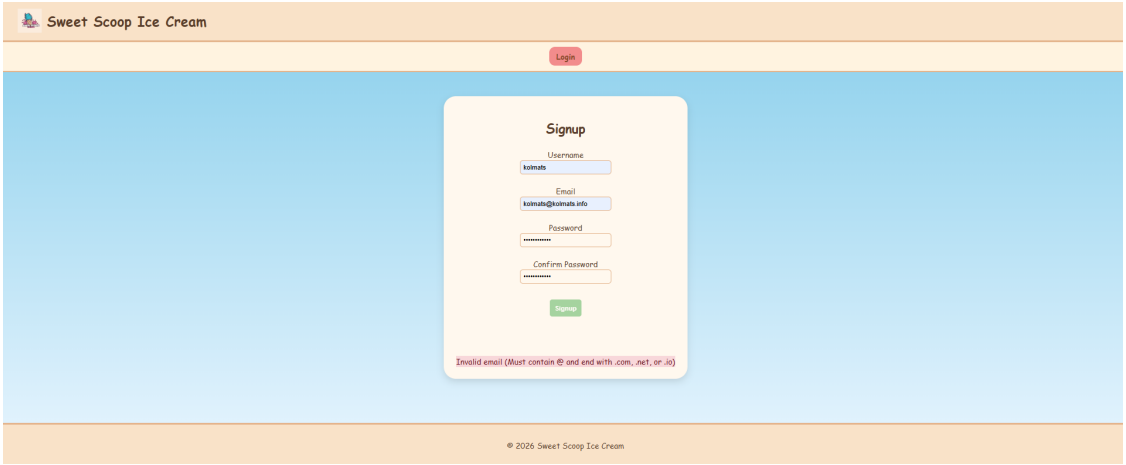
- Must contain an “@” symbol followed by a domain name (.com, .net, .io).

c) ***Signup Button Behavior****:*

When the user clicks on the “Sign up” button, perform the validation and:

- *Display error messages* in a dynamic box for invalid inputs.
- On *successful signup*, display a success message in a dynamic box and then redirect the user to the Login page.





Sweet Scoop Ice Cream

Login

Signup

Username

conrad

Email

conrad@conard.com

Password

••••••••••••

Confirm Password

••••••••••••

Signup

Signup successfull Redirecting to login...

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